

1. Which algorithm is used in the Game tree to make decisions of Win/Lose?
 - A. DFS/BFS Algorithm
 - B. Greedy Search Algorithm
 - C. Heuristic Search Algorithm
 - ☒ D. Min/Max Algorithm

2. Which of the following people should use a heuristic?
 - A. Siti, who is designing a program to solve a problem with a small, finite number of solutions 不需要
 - B. Jason, who is designing a program to solve a problem with no solution —做不到
 - C. Ramesh, who is designing a program to solve a problem that requires little time
 - ☒ D. Mei Ling, who is designing a program to solve a problem that requires little processing power

3. Which of the following is **TRUE** about informed heuristic?
 - A. An informed heuristic does not ignore any child node thus will update open and closed list in order to find shortest path the first time the node is visited ✗
 - B. A heuristic $h(n)$ is informed if it guarantees to find actual cost path to a solution
 - ☒ C. $h(n)$ is more informed when $h(n) \leq h^*(n)$ ✓
 - D. When a heuristic $h(n)$ is more informed this means that the heuristic evaluates less state space to expand to return optimal solution

4. What is the primary goal of the **Minimax algorithm** in game theory?
 - A. To maximize the number of moves for both players.
 - B. To predict the opponent's moves accurately.
 - ☒ C. A decision-making algorithm widely used in two-player game.
 - D. To explore all possible moves and their outcomes without considering the opponent.

5. In the Minimax algorithm, what does a "MAX" node represent?
 - A. A node where the minimizing player chooses their move.
 - ☒ B. A node where the maximizing player chooses their move.
 - C. A node where no player makes a decision.
 - D. A terminal node in the game tree.

6. Which of the following best describes Alpha-Beta Pruning?
- A. A method to expand the game tree to deeper levels.
 - ☒ B. A technique to eliminate branches that won't influence the final decision.
 - C. A strategy to ensure the game always ends in a tie.
 - D. An algorithm that skips evaluating terminal nodes.
7. How does Alpha-Beta Pruning enhance the Minimax algorithm?
- ☒ A. By reducing the time complexity of node evaluation.
 - B. By increasing the branching factor of the game tree.
 - C. By skipping leaf nodes entirely.
 - D. By always exploring the entire game tree for better results.
8. Alpha-Beta pruning is applied in which type of games?
- A. Single-player games
 - ☒ B. Information games like Chess or Checkers
 - C. Probabilistic games like Snakes and Ladders
 - D. Games with hidden information
9. What is the role of the knowledge base in a production system?
- ☒ A. Storing rules and facts
 - B. Managing user interface
 - C. Compiling programming code
 - D. Optimizing database queries
10. A production system has conflicting rules that can fire simultaneously. Which strategy would best resolve the conflict?
- A. Rule elimination
 - ☒ B. Conflict resolution using priority or specificity
 - C. Random selection
 - D. Adding more rules

11. Given the production system rule: IF a person has a fever AND a sore throat, THEN they might have the flu. What would happen if there's no sore throat but a fever and muscle pain instead?

- A. The rule will still fire.
- ☒ B. The rule will not fire.
- C. The system will generate a new rule.
- D. The system will terminate.

12. What is the basic structure of a production rule?

- ☒ A. Condition $\rightarrow$ Action
- B. Action $\rightarrow$ Condition
- C. Rule $\rightarrow$ Conclusion
- D. Statement $\rightarrow$ Explanation

13. In an expert system that uses production rules, how can you improve the accuracy of the inferences made by the system?

- A. Add more rules, even if they contradict each other
- B. Use only the most recent rules and discard older ones
- ☒ C. Regularly update the knowledge base and ensure rules are accurate and well-prioritized
- D. Use simpler rules without conditions to speed up decision-making

14. What is the primary function of an intelligent agent?

- ☒ A. To perform tasks on behalf of a user
- B. To store large amounts of data
- C. To enhance hardware performance
- D. To provide network security

15. Which type of AI agent is most suitable for dynamic and unpredictable environments?

- A. Simple reflex agents
- B. Model-based reflex agents
- C. Goal-based agents
- ☒ D. Utility-based agents

16. In AI, the term “rational agent” refers to an agent that:

- A. Always follows human commands
- ☒ B. Acts to achieve the best outcome
- C. Has a physical presence
- D. Requires minimal resources

17. Consider an automated online shopping agent that assists users in purchasing books. Which PEAS description is most appropriate?

- A. **Performance Measure:** Number of books displayed; **Environment:** User’s computer; **Actuators:** Monitor display; **Sensors:** Keyboard input.
- ☒ B. **Performance Measure:** User satisfaction and cost minimization; **Environment:** Online bookstores; **Actuators:** HTTP requests to purchase books; **Sensors:** Web scraping tools to gather book information.
- C. **Performance Measure:** Speed of search; **Environment:** Local database; **Actuators:** SQL queries; **Sensors:** Database triggers.
- D. **Performance Measure:** Number of users; **Environment:** Social media platforms; **Actuators:** Posts and messages; **Sensors:** User interactions.

18. An online shopping assistant helps users find and purchase products. Which type of AI agent best describes this system?

- A. **Simple Reflex Agent:** Acts based on current perceptions without considering past experiences.
- B. **Model-Based Reflex Agent:** Uses internal models to handle partially observable environments.
- C. **Goal-Based Agent:** Acts to achieve specific objectives, such as completing a purchase.
- ☒ D. **Utility-Based Agent:** Evaluates different actions based on a utility function to maximize satisfaction.

19. Which of the following examples best reflects the concept of clustering?

- I. Predict customer buying preferences based on their interaction history and purchasing behaviour.

II. Identifying patterns in patient data to determine who will respond better to a specific treatment.

classification

III. Sorting emails into different folders based on their content. ~~X~~ 有 label

IV. Forecasting book demand in a library by analysing borrowing patterns and reader preferences. ~~X~~ 有 label

A. I and II

B. **II and III** ~~X~~

C. I, II, and IV ~~X~~

D. I, III, and IV ~~X~~

label data

20. Which of the following is valid about supervised learning?

~~I.~~ Predicting house prices based on features like size, location, and number of bedrooms.

II. Predicting future outcomes based on historical data.

~~III.~~ Training an agent to play a game by rewarding it for good moves and penalising it for bad ones.

~~IV.~~ Classifying emails as spam or not spam. ✓

A. I and II ~~X~~

B. II and III ~~X~~

C. **I, II, and IV**

D. I, II, III, and IV

21. Which of the following is true about machine learning?

A. Regression is a type of supervised learning, while association is a type of unsupervised learning.

B. Both reinforcement learning and supervised learning use external supervision for training. ~~X~~

C. Both reinforcement learning and unsupervised learning use unlabelled data.

D. K-Means algorithm maps labelled input to known output while the Support Vector Machine understands patterns and discovers output.

22. Which of the following data is the most valid to forecast house prices?

A. Age of the property and historical house prices.

B. Age of the property and category of house prices.

- C. Historical house prices and category of house prices.
- D. Distance to commercial centres and workplaces and category of house prices. ✗

23. A retail company wants to segment its customers based on purchasing behaviour to design personalized marketing campaigns. Which technique should they use?

- A. Classification
- ☒ B. Clustering
- C. Regression
- D. Reinforcement Learning

24. A healthcare provider builds a model to classify patients as "High Risk" or "Low Risk" for a certain disease. They want to ensure they do not miss any high-risk patients. Which metric should they focus on?

- A. Precision
- ☒ B. Recall
- C. Mean Absolute Error
- D. R-Squared

25. A spam detection system classifies emails as spam (positive) or not spam (negative). During evaluation, the confusion matrix shows a high number of false positives. What could be a potential impact?

- ☒ A. Genuine emails being marked as spam
- B. Spam emails being delivered to the inbox
- C. Lower precision ✗
- D. Higher recall ✗

B. Structure Question: 75 Marks

Question 1 (15 marks)

Figure 1 below illustrates a search graph from the start state S, to goal G. A, B, C and D are the states that can be traversed from A to G. Answer all the following questions based on the figure 1 below.

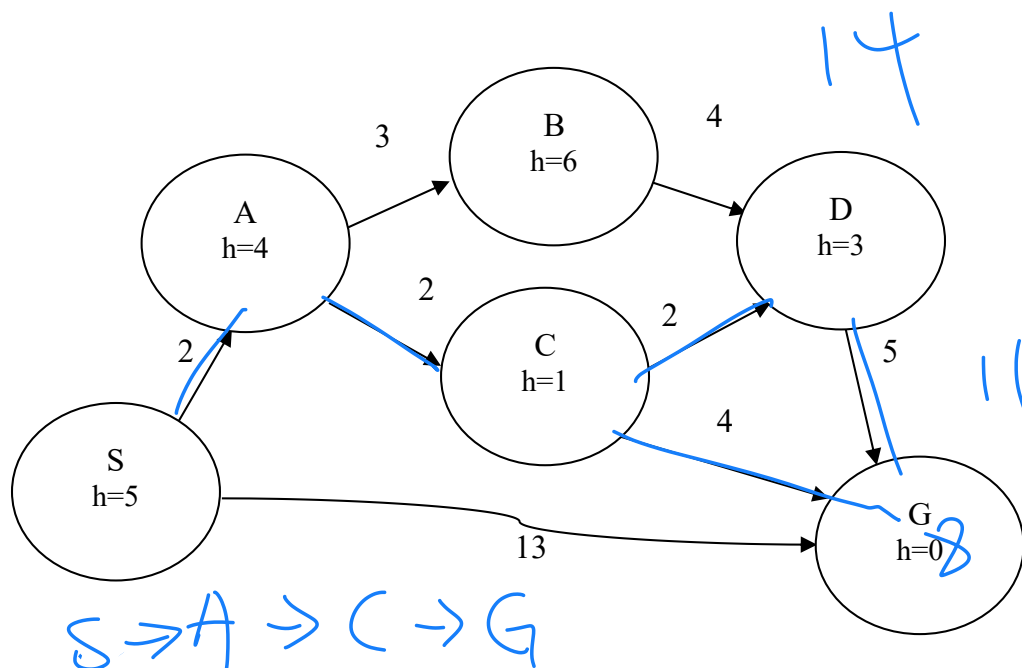


Figure 1. A search graphs

- a. Using Algorithm, A* evaluate whether the optimal path from S to G is consistent in the table 1 given below [9m]

Table 1. Heuristic evaluation

n-n'	h(n)	h(n')	c(n-n')	h(n')+c(n-n')	
S-A	5	4	2	6	✓
S-G	5	0	13	13	✓
A-B	4	6	3	9	✓
A-C	4	1	2	3	✗
B-D	6	3	4	7	✓
C-D	1	3	2	5	✓
C-G	1	0	4	4	✓
D-G	3	0	5	5	✓

- b. From your answer in 1a), is the heuristic consistent? Explain how is the heuristic is greater than consistent / not consistent. Justify your answer by giving example [3m]
- For the path A-C the $h(A)=4$, $h(C)=1$, $c(A,C)=2$. $4 > 1+2$. the consistency condition is violated.
- Not consistent. The condition for consistent is $h(n) \leq h(n') + c(n,n')$.
- c. Consider that each state in the graph above are cities in Malaysia with each path showing the distance between each city and the heuristic $h(n)$ is the straight-line distance between the city n and the destination city G . Given that the optimal solution path is S-A-C-G,

$$\begin{array}{ll}
 S: h(S) \leq h^*(S) : 5 \leq 8 \checkmark & G: h(G) \leq h^*(G) : 0 \leq 0 \checkmark \\
 A: h(A) \leq h^*(A) : 4 \leq 6 \checkmark & D: h(D) \leq h^*(D) : 3 \leq 5 \checkmark \\
 C: h(C) \leq h^*(C) : 1 \leq 4 \checkmark & B: h(B) \leq h^*(B) : 6 \leq 9 \checkmark
 \end{array}$$

it is admissible. since $h(n)$ is always less than or equal to the cost $h^*(n)$ for all node.
 explain whether the heuristic $h(n)$ is admissible / not admissible for a person traveling from city S to city G. Justify your answer by giving example [3m]

Question 2 (5 marks)

Consider the following game tree in Figure 2, where Player 1 aims to maximize their score. The numbers at the leaf nodes represent the values provided by a static evaluator for the corresponding board positions.

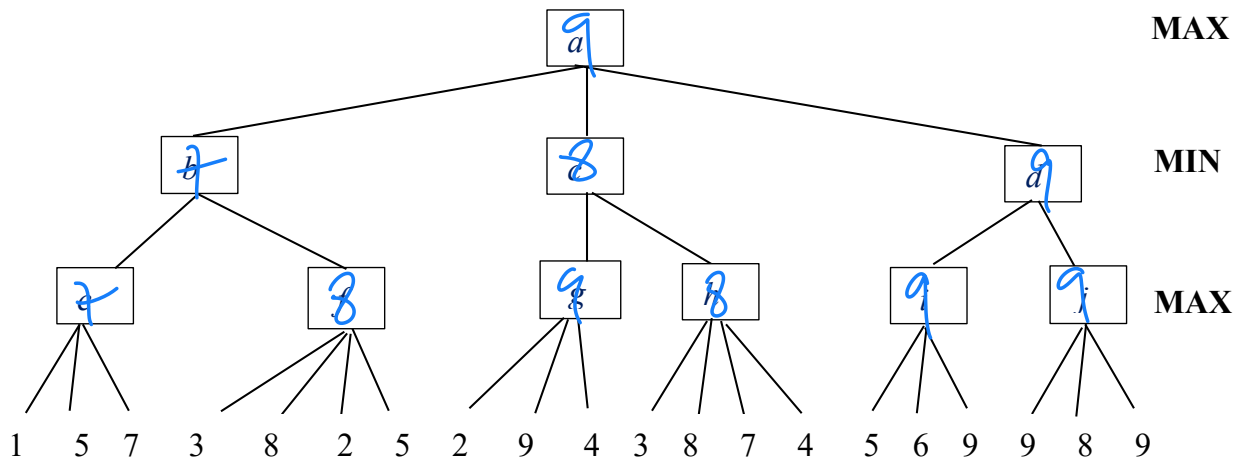


Figure 2: Game tree using Minimax

Using the standard Minimax algorithm, determine the values of nodes labelled (a to j) in the game tree. (5 marks)

Question 3 (15 marks)

In the game tree in Figure 3 below, it is Max's turn to move. At each leaf node is the estimated score of that resulting position as returned by the heuristic static evaluator.

Analyse and calculate the values for the following questions:

- Perform Mini-Max with alpha beta pruning search and calculate the value of node A, B, C, D, E, F, G, H, and I. $A = 11$ (8 marks)
- Examine and select the nodes that will not be evaluated. O, H, S (3 marks)
- Calculate the α and β final values for node F? $\alpha = 12, \beta = 11$ (2 marks)
- Calculate the optimal value for MAX to win the game? 11 (1 mark)
- Explain why Alpha-Beta Pruning improves the efficiency of evaluating the tree compared to the basic Minimax algorithm. (1 mark)

it reduce the time complexity of the node evaluation.

$\max : \alpha = -\infty$
 $\min : \beta = \infty$

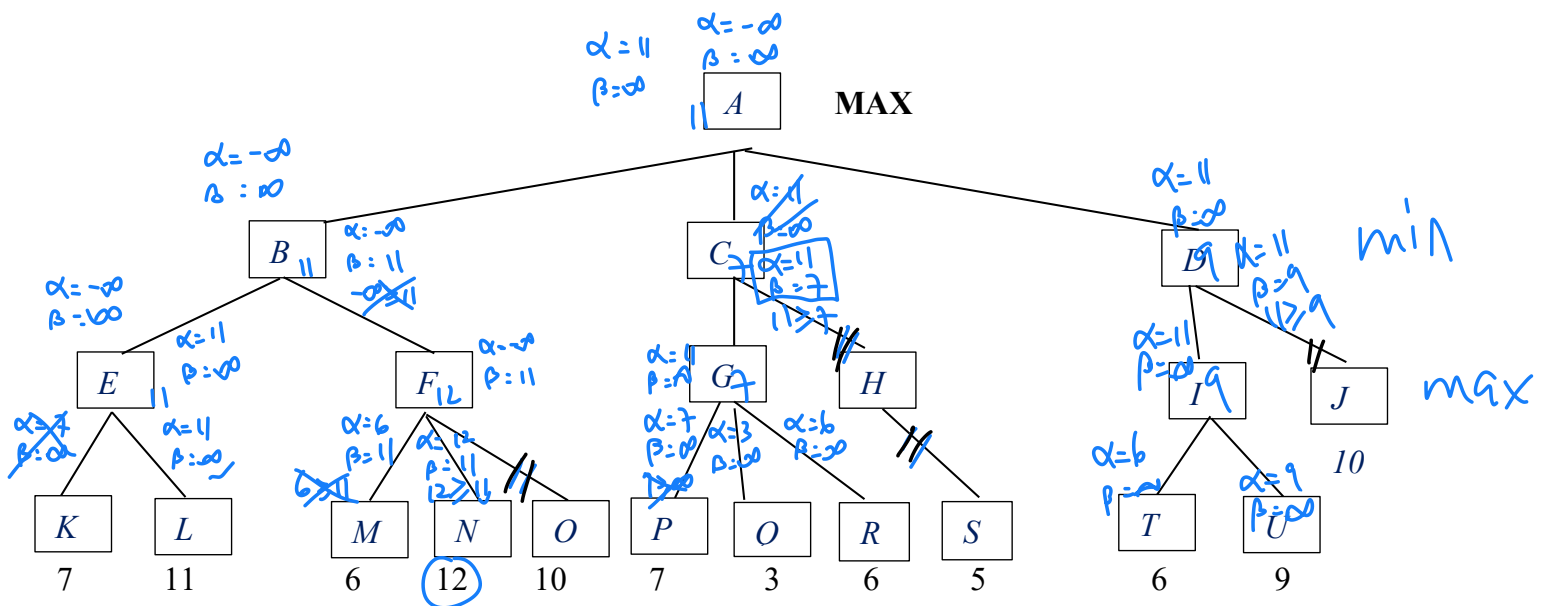


Figure 3: Game tree using Alpha-beta Pruning

Question 4 (14 marks)

A production system is being used to identify potential causes of a system malfunction in a factory. The system uses the following rules in its **knowledge base**:

Rule 1: IF machine temperature $> 80^{\circ}\text{C}$, THEN trigger "Overheating alert".

Rule 2: IF "Overheating alert" AND coolant level $< 50\%$, THEN diagnose "Coolant issue".

Rule 3: IF "Overheating alert" AND fan speed < 60 RPM, THEN diagnose "Fan issue".

Rule 4: IF "Overheating alert" AND power supply fluctuation = TRUE, THEN diagnose "Power supply issue".

At the beginning of the process, the **working memory** contains the following facts:

- Machine temperature = 85°C **A**
- Coolant level = 30% **C**
- Fan speed = 50 RPM **E**
- Power supply fluctuation = TRUE **G**

Using the provided scenario, answer the following:

- Fill in the table 2 below for **three iterations** of the production system, including:
 - The current **working memory**.
 - The current **conflict set** (rules that match the current working memory).
 - The rule selected and fired in the current iteration.
 - The updated working memory after the rule is fired.

Table 2. Production system

Iteration	Working Memory	Conflict Set	Rule Fired	Updated Working Memory (Output)
1	A, C, E, G	1	1	A, C, E, G, B
2	A, C, E, G, B	1, 2, 3, 4	2	A, C, E, G, B, D
3	A, C, E, G, B, D	1, 2, 3, 4	3	A, C, E, G, B, D, F

(12 marks)

- b. Explain how the system resolves conflicts when multiple rules are applicable at the same time, if any. (2 marks)

- use first match that does not lead a loop
- use priority - most important rule is selected to fire when multiple rule match the working memory.

Question 5 (10 marks)

- a. Define an intelligent agent. Provide **TWO (2)** examples of where intelligent agents are used. (4 marks)

Intelligent agent is an agent that can be viewed as preceiving its environment through sensor and acting upon that environment through actuators.

robotic vacuum cleaner - siri / autonomous vehicle

UTM has developed an AI system (Mental Health Detection) to identify students who may be experiencing mental health challenges. The system analyzes data from student interactions on the university's online platforms, such as forum posts and messages, to detect signs of stress, anxiety, or depression. When potential issues are identified, the system alerts university counselors (Pusat Kaunseling UTM) to provide appropriate support.

Based on the case study, answer all questions:

- b. Describe the PEAS components for the Mental Health Detection system.

[4 marks]

- Performance Measure: alert counselors, accuracy of detecting student at risk,
 - Environment: university's online platforms, forum post and messages.
 student interact through
 - Actuators: alert counselors
 - Sensors: student forum post, message, text data
- c. Choose the most suitable type of AI agent for the Mental Health Detection System and give your justification. [2 marks]

Question 6 (15 marks)

The MeowScanner can identify whether a cat is a domestic cat (L) or a Persian breed (P). Table 3 showcases the cat dataset, which includes characteristics such as the length of the nose, fur, and tail, as well as the size of the eyeballs. [10 marks]

Table 3: Cats dataset

Actual Class	Nose Length	Fur Length	Eyeball Size	Tail Length	Predicted Class
L	Normal	Short	Proportional	Standard	L ✓
P	Short	Long	Large	Bushy	P ✓
L	Normal	Medium	Proportional	Standard	L ✓
P	Short	Long	Large	Bushy	L ✓
L	Normal	Short	Proportional	Standard	P ✓
P	Short	Long	Large	Bushy	P ✓
L	Normal	Medium	Proportional	Standard	L ✓
P	Short	Long	Large	Bushy	L ✓
L	Normal	Short	Proportional	Standard	L ✓
P	Short	Long	Large	Bushy	P ✓
L	Normal	Medium	Proportional	Standard	P ✓
P	Short	Long	Large	Bushy	L ✓
L	Normal	Short	Proportional	Standard	L ✓
P	Short	Long	Large	Bushy	P ✓
L	Normal	Medium	Proportional	Standard	L ✓

Handwritten Confusion Matrix:

	Predict P	Predict L	
Actual P	4 TP	3 FN	
Actual L	2 FP	6 TN	

Based on Table 3, answer the following questions.

- Quantify the number of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) for the classification result [4 marks]
- Using your answers in a), calculate the accuracy, precision and recall for this classification. [6 marks]
- Discuss whether the classification model is effective for detecting breed of cat, considering the trade-offs between precision and recall. (3 marks)

Handwritten notes for question c):
 recall = 0.57, precision = 0.67, low recall, high precision, this shows the classification is not effective, shows that we miss a lot of positive examples.

- d. Suggest one classification model that appropriate to solve the above problems and justify (2 marks). *Decision tree, it use logic rule to explaining the classification*

$$b. \text{ accuracy} = \frac{TP+TN}{(TP+TN+FP+FN)}$$
$$= \frac{10}{15}$$

$$= 0.67$$
$$\text{precision} = \frac{TP}{(TP+FP)}$$
$$= \frac{4}{(4+2)}$$

$$= 0.67$$
$$\text{recall} = \frac{TP}{(TP+FN)}$$
$$= \frac{4}{(4+3)}$$
$$= 0.57$$